

Average Rate of Change



Computing average rate of change

- 1 Find the average rate of change of $f(x) = x^2$ over each interval:
a $[0, 2]$ **b** $[1, 3]$ **c** $[2, 5]$ **d** $[-1, 1]$
- 2 Find the average rate of change of $f(x) = 2^x$ over the interval $[0, 3]$.
- 3 Show that the average rate of change of $g(x) = 3x - 5$ is the same over any interval $[a, b]$, and state its value.

Applications

- 4 A ball is thrown upward so that its height after t seconds is $h(t) = -5t^2 + 20t$ meters.
 - a** Find the average rate of change of the height over $[0, 2]$.
 - b** Find the average rate of change of the height over $[1, 3]$.
 - c** Interpret your answer to part b in the context of the ball's motion.
- 5 The graph of $f(x) = -\frac{1}{2}x^2 + 2x + 1$ is shown. Estimate the average rate of change over $[0, 4]$, then verify your estimate algebraically.

